Hsuan-Fu (Jeff) Hsu

AI and ML Engineering | Model Evaluation, Retrieval and LLM Agents

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EDUCATION

University of Southern California — Los Angeles, CA
Master of Science in Computer Science | GPA: 3.6/4.0 — Jan 2026 - May 2027 (Expected)

National Taipei University — New Taipei City, Taiwan
Bachelor of Science in Computer Science | GPA: 3.5/4.0 — Sept 2020 - June 2024

SKILLS

Languages: Python, TypeScript, C++, C#, SQL

ML and Evaluation: PyTorch, Sentence Transformers, TensorFlow, NumPy, tsfresh, LOSO validation

Agents and APIs: LangChain, LangGraph, MCP, OpenAI Agents SDK, FastAPI

Cloud and Testing: Docker, Google Cloud Run, pytest, Vitest, xUnit, GitHub Actions, Git

EXPERIENCE

National Taipei University — New Taipei City, Taiwan
Undergraduate Researcher | Advisor: Prof. Bor-Shing Lin — July 2022 - Feb 2024

- Built a **Python** pipeline for noninvasive uric acid estimation from 30 participants and two wavelengths, applying Bessel filtering, quality checks, and tsfresh feature extraction.
- Compared five ML model families and four DL architectures using **LOSO cross validation**; the team's linear regression model achieved 10.3% MAPE with 10 selected features.
- Received individual NSTC undergraduate research funding and earned fourth place in NTPU's project competition with a team of three.

Tech-Management Workshop Co., Ltd. — Taichung, Taiwan
Software Engineer Intern — July 2026 - Aug 2026

- Independently built a government application with **ASP.NET Core MVC** and **SQL Server**, deployed across 29 offices with status tracking and incremental data synchronization.
- Reduced database latency by at least 50% and overall resource usage by at least 30% through data processing and storage improvements.

PROJECTS

Scientific Retrieval Fine-tuning — Personal Fork — Oct 2026
Python, PyTorch, Sentence Transformers | GitHub

- Fine-tuned **MiniLM** with **PyTorch** and Sentence Transformers, improving mean nDCG@10 by 3.2% and Recall@10 by 2.1% over random-negative training on 300 SciFact test claims across three seeds.
- Built a controlled evaluation workflow with data leakage checks, false-negative review, 53 unit tests, and model save/reload verification.
- Extended the fine-tuned retriever into a **FastAPI** retrieval-augmented QA service with FAISS search and cited answers from a local LLM (Ollama), reproducing the original nDCG@10 and Recall@10 exactly on 300 test claims for all 4 models.

Promptfoo Agent Retry Validation — Personal Fork — Oct 2026
TypeScript, Vitest, OpenAI Agents SDK | GitHub

- Added nested retry policy error paths using **TypeScript**, with six regression cases failing before the change and passing after it.
- Added 81 test cases with **Vitest** and the OpenAI Agents SDK, including 60 configuration checks and 11 retry behavior scenarios with mocked model clients.

LLM Agent with MCP Tools — Aug 2026 - Present
Python, LangGraph, FastAPI, Google Cloud Run

- Built a **LangGraph** agent integrating three MCP servers, session memory, tool retries, and per-session locks within a service instance.
- Deployed a **FastAPI** service with Docker on Google Cloud Run and added pytest tests for tool validation and error handling.

PUBLICATION

Coauthor, "Noninvasive Portable System With Dual Near-Infrared Light Technology for Monitoring Uric Acid Concentration," IEEE Transactions on Human-Machine Systems, accepted for publication, Aug 2026.